

EMILY DEEB

Architect, Urban Designer & Computational Analyst | MSc Urban Spatial Science, CASA UCL
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PROFESSIONAL SUMMARY

Architect and urban designer with five years of professional practice at city scale, combining strategic design leadership with geo-computational analysis. Led teams across 15 concurrent projects for municipal and private clients, developing GIS workflows, market analysis tools, and spatial data pipelines that improved office efficiency by 30%. Now completed a degree in Urban Spatial Science at CASA UCL, with an industry dissertation applying Python-based machine learning to national retail spending datasets in collaboration with CACI Ltd. Bringing a rare combination of design intelligence, spatial analytical rigour, and the ability to communicate complex findings to non-technical stakeholders.

EDUCATION

- 2025 – 2026* **MSc Urban Spatial Science** | The Centre for Advanced Spatial Analysis (CASA), UCL
- Specialising in computational urban analysis, spatial data science, geocomputation, statistical modelling, urban simulation, data visualisation, and machine learning.
- 2015 – 2020* **B.Arch. Architecture** | Bezalel Academy of Arts and Design Jerusalem
- Graduation Project: Destination: Jericho — research-led urban design study using QGIS, spatial analysis, and data visualisation. Nominated for David Azrieli Architecture Prize.

PROFESSIONAL EXPERIENCE

- June 2026* **Buro Happold, London** | Industry Consultant — Building a Better Future Scheme
- Led research scoping and team coordination across a multidisciplinary group of six; managed stakeholder expectations, synthesised meeting outputs, and contributed to the final written report and client presentation delivered to Buro Happold and industry stakeholders.
- 2020 – 2025* **Derman Verbakel Architecture, Tel Aviv** | Team Leader — Urban Design & GIS
- Led a team of 4 architects across 15 concurrent projects spanning block to city scale for municipal and private clients including Eilat Municipality and the Israeli Authority for Urban Renewal.
 - Developed and led geo-computational workflows in QGIS and ArcGIS for morphological mapping, site feasibility analysis, market analysis, and statistical profitability modelling; improving office efficiency by 30%.
 - Independently collected, cleaned, and aggregated spatial datasets from scratch for Sakhnin city centre regeneration; delivered evidence-based spatial reports and cartographic visualisations to municipal planning committees.
 - Trained 15+ team members in GIS tools; developed reproducible, documented spatial workflows adopted across the office.
 - Key projects: Transportation Masterplan of Eilat; Regeneration of Sakhnin City Centre (1,200 units); TOD Regeneration Plan Rishon LeTsiyon (3,000 units); City expansion masterplan (100,000 units, confidential).
- 2023-2025* **Bezalel Academy of Arts and Design, Jerusalem** | Professor Assistant

- Co-taught second-year B.Arch design studio, mentoring twenty students through design critique, spatial analysis workshops, and individual guidance.

2019 – 2020

SYL Studio Yigal Levi Architecture, Jerusalem | Architect Intern

- Gained foundational experience in architectural design, documentation, and site coordination.

SELECTED PROJECTS

For full portfolio: <https://emilydeeb.com/>

2026

Retail Spending & Urban Morphology Prediction | Industry Dissertation

CACI Ltd x GeoDS Programme

- Investigated whether urban morphological fabric type predicts retail spending patterns across UK cities, independently of demographic composition, using CACI proprietary consumer spending datasets and HiMoC morphometric classifications.
- Built Python-based machine learning pipeline using GeoPandas, scikit-learn, SHAP, and spatial autocorrelation analysis; applied pre-modelling diagnostics and statistical modelling to produce reproducible, documented analytical outputs.
- Communicated findings through data visualisation and written reporting to CACI's Consumer and Location Intelligence team; presented at GeoDS Data Partner Forum, Saïd Business School, Oxford.

2023

TOD Regeneration Plan, Future Metro Station | Derman Verbakel Architecture

Client: Israeli Authority for Urban Renewal | 3,000 units, Rishon LeTsiyon

- Led design and spatial analysis for a Transit-Oriented Development regeneration plan surrounding a future metro station, covering 3,000 residential units across mixed-use urban blocks.
- Developed geo-computational workflows in QGIS and ArcGIS for morphological mapping, profitability modelling, and statutory planning submissions to the Israeli Authority for Urban Renewal.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Matplotlib, GeoPandas, Scikit-learn), R (RStudio), SQL, React.

Data Science & ML: Classification, Clustering, Regression, Statistical Analysis, SHAP, Spatial Autocorrelation, Pre-modelling Diagnostics, Predictive Modelling

Data & Viz: Data Pipelines, Reproducible Workflows, Data Visualisation, Power BI, Quarto, Jupyter, Git

Spatial & GIS: QGIS, ArcGIS, GeoPandas, Spatial Network Analysis, Spatial Data Science, Urban Morphology, Google Earth Engine API

Design & Other: AutoCAD, Revit, Rhinoceros, Adobe Suite (Photoshop, Illustrator, InDesign), Microsoft Office, Excel, UI/UX (CalArts certified)

LANGUAGES & CERTIFICATIONS

English (fluent), Hebrew (fluent), Arabic (native)

Licensed Architect — Israeli Labor Department, 2025